

K-808037

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA07

COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K-808037 192525035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K-808037
Signature of the student:

=> MAC address identifies devices within the same local network

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ASSESSMENT TOOL-2

①

OSI Layer:-

Application



Presentation



Session



Transport



Network



Data Link



Physical

Application layer



Presentation layer



Session layer



Transport layer



Network layer



Data Link layer



Physical layer

missing OSI layer:-

=> Establishes, manages and terminates communication sessions

=> maintains synchronization between sender and receiver.

Data Link layer:-

=> provides error detection & flow control

=> uses MAC addresses for local delivery and converts packets into frames

Application layer

↓

Transport layer

↓

Network layer

↓

Data link layer
(frames)

↓

Physical layer
(bits)

Analysis:

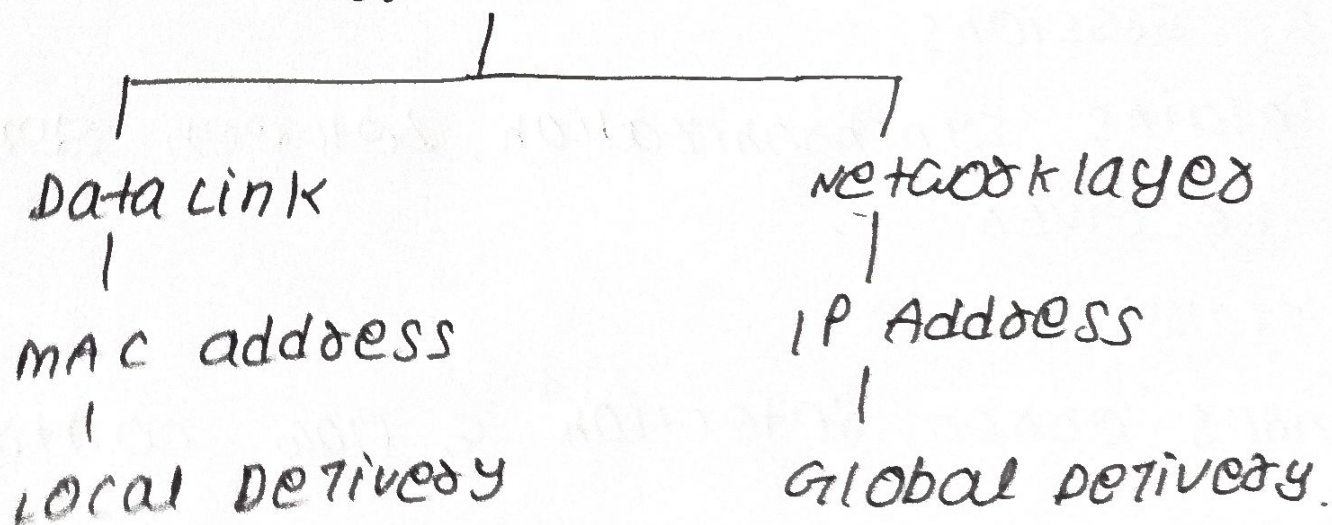
⇒ Transport layer divides data into segments

⇒ Network layer adds IP information to create packets.

⇒ Data link layer adds MAC addresses and error checking to form frames

MAC Address & IP Address:

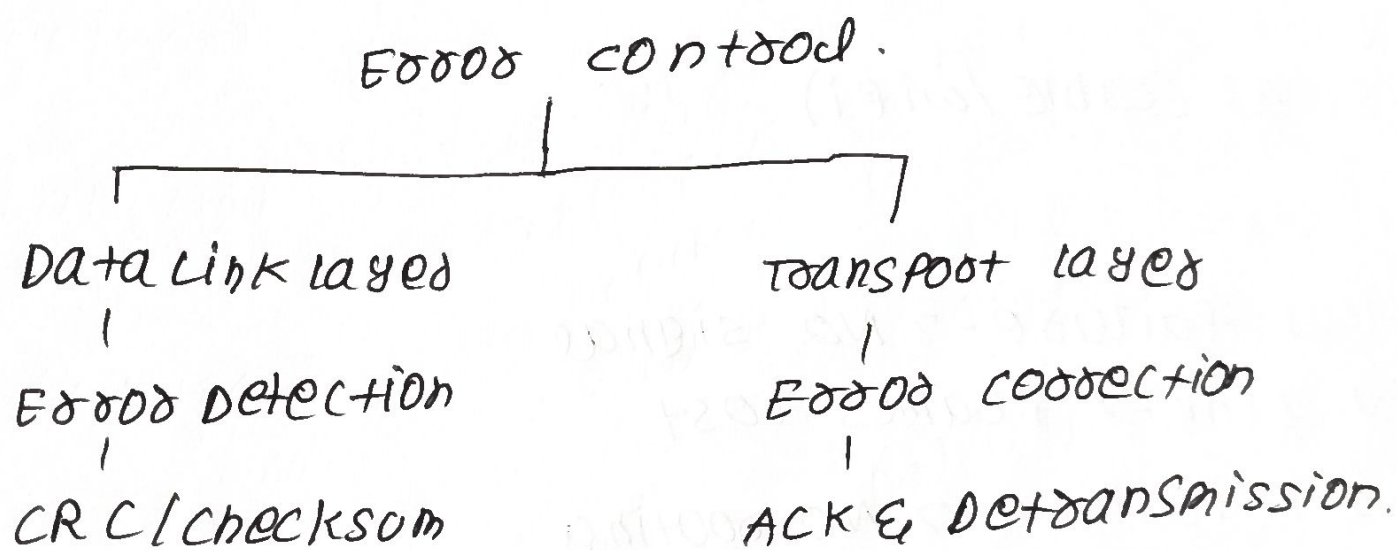
Data Delivery



Explanation:-

- => MAC address identifies devices within the same local network
- => IP address identifies devices across different networks
- => Data Link layer device frames locally using MAC addresses
- => Network layer routes packets globally using IP addresses.

④ Error control:-



Analysis:-

- => Data Link layer detects transmission
- => Transport layer corrects errors by requesting retransmission.
- => Together, they improve reliability accuracy and successful data delivery

⑤ Web Browsing using OSI Layers:

Web Browsers (HTTP)



Presentation (SSL/TLS)



Session



Transport



Network



Data Link (MAC)



Physical (cable/wifi)

Analysis:-

Physical failure → No signal

Data Link → Frames lost

Network failure → No routing

Transport failure → connection breaks.

Application failure → web pages cannot load

⑥ Network Issue Analysis

NETWORK TROUBLESHOOTING



Physical



Data Link



Transport



Application → Network layer failure.